



Smart lighting components - Philips Lighting InterAct series

taking centrestage. In this article we discuss how these technologies can benefit small- and medium-scale offices, and what will be the ROI for them.

Benefits and latest solutions

Until recently, small- and medium-sized workplaces opted for generic lighting setups, looking at lower capex requirement. They wanted to avoid risk of low ROI. Lack of advanced lighting solutions with a strong business value and lack of right advice on lighting for a small office space also factored in. In turn, they ended up spending much more on electricity with no way to customise their lighting to enhance employee comfort.

Today, businesses are recognising the importance of employee experience in improving productivity. Personalisation of the workplace can be a major motivator, and smart lighting solutions can be a capable agent.

A Gurugram-based smart home solutions startup comments on its investment in smart lighting, “With options in lighting colours, employees can set the ambience as per their needs. This helps them become comfortable in their workspace, which, in turn, motivates them to perform better. After long hours of work, if they feel like relaxing, they can dim the lights accordingly.”

This benefit is derived from the capability of smart lighting solutions to mimic natural daylight and promote a healthy indoor ambience—a concept called human-centric lighting.

Sensor-based lights and dimmers can reduce electricity wastage by up to 25 per cent. Mahesh Bellad, director, Online Instruments (India) Pvt Ltd, says, “With the help of simple sen-

sors across a workplace, we can control a group of lights and, thus, provide additional cost savings on top of what the LED lighting in general provides.”

A small office owner in Bengaluru says, “Smart lights powered by sensors are highly useful for individual office cabins or conference rooms. These can power on when someone walks in, and turn off on exiting, ensuring that

the lights are optimally used. Similar applications along washrooms and less-used corridors are also possible.”

Smart lights have been modernised with wireless lighting solutions. Wiring and cables not only bring heavy installation hassles and costs, but also make it difficult for organisations to move their lighting infrastructure when needed. Wireless solutions, on the other hand, reduce installation and maintenance costs.

Philips Lighting has developed InterAct Series of smart lighting, focussed on small and medium offices. These are available in various lighting form factors, like 2x2 luminaires, downlighters, spot lighting and TLEDs. With the help of a wireless network gateway (InterAct gateway), luminaires can be configured and controlled using an app (InterAct Pro) or over the Web.

InterAct Pro dashboard delivers insights on real-time energy usage and various consumption patterns. Lighting sensors automatically dim or power off the lights based on occupancy. Moreover, occupants can remotely control the lights to set a personalised ambience in offices.

With real-time energy consumption measurement and centralised dashboard-based insight generation, smart lighting solutions allow workplaces to optimise their lighting usage. This, in turn, reduces electricity expenses. Moreover, scheduled lighting provides an additional sense of convenience, as office owners can power on or off their lights following specific timings.

Price points and ROI

Initial investments in smart lighting may be slightly higher as compared to

normal solutions, depending on scale. However, in a well-planned setup, recovery of investments can be quick.

Sumit Joshi, vice chairman and managing director, Philips Lighting India, says, “Cost uplift for smart lighting can be anywhere between 1.5 and 1.7 times compared to conventional lights, depending on the number of light points. But the comparison should be done based on ROI. ROI should be calculated after adding the benefits of various features like occupancy, daylight harvesting, scheduling, zoning, dashboards and personalisation to base energy savings through LEDs.

“Keeping this in mind, the usual ROI of two years in a non-intelligent lighting system becomes around three years for the intelligent wireless lighting solution, with better returns going forward.”

Bellad confirms, “Prices are flexible as features and scalability can be enhanced or reduced as per customer requirements. In the cases we have seen, ROI was obtained within 15 to 18 months, subject to variations in usage patterns.”

With a clear picture of all technologies and expected benefits, offices can start investing in innovative lighting solutions with long-term profits in mind.

—Paromik Chakraborty, technical journalist, EFY

The IoT In Hospitality: Advantages And Limitations

➤ With proper implementation, the IoT can be an effective tool to increase customer satisfaction and, consequently, increase business revenue